Gretchen Oracle Project Report

Abstract:

Deep within the urban landscapes that we are all ever so used to, there is a world of mystery and divination. Tarot cards have been a tool that has been used for centuries, evolving from a trick-taking card game called Tarocchi in Northern Italy to instruments of fortune telling in France. Whether one believes in the power of Tarot, its charm remains unmatched against the testament of time. Interpreting the cards is an art form that requires a tremendous amount of patience and experience to be able to do it on its own. As a group we wanted to see if we were able to make Gretchen interpret the Tarot cards for you so that we will be able to make Tarot readings more accessible to the general public.

Introduction and Project Goals:

The objective of the project is to read the tarot cards to deliver a proper prediction for any individual.

1. Individuals will be given the choice of choosing between the following predictions
    a. Shhh… (A secret surprise)
    b. Personality
    c. Career
    d. Love
    e. Yes/No
2. We created a dataset which contains pictures of all the required cards. We will use this dataset to train our model to recognise any card that has been shown to Gretchen.
3. We will be using YOLO for Gretchen to be able to identify the card that is being shown. It will then be able to match the card to answer the question that has been asked by the user.
4. Any permutation and combination of the cards will be able to receive a prediction as we will be using Artificial Intelligence to interpret the cards that have been shown to Gretchen.
5. In order to make the experience more immersive, we will be adding music before and during the prediction. Predictions will be delivered using text as well as text-to-speech.
6. In order to end the reading users will need to close their hands in the shape of a fist to allow for everything to be reset.

Technologies used:

1. YOLO for card recognition
    a. Based on our own dataset, we were able to train Gretchen to be able to recognise the card that is being presented to the camera.
2. MediaPipe for gesture recognition
    a. Gretchen is able to identify our hand gestures in order to set the mode of the reading.
    b. Showing a fist will exit the current mode.
    c. Showing one finger will set the mode to “Shhh…”, which plays the music and the video of “Never Gonna Give You Up” by Rick Astley. In tarot, it is said that each deck contains the energy of the owner. Therefore, we decided to leave a piece of ourselves in this mystery mode.
    d. Showing two fingers will set the mode to “Personality”.
    e. Showing three fingers will set the mode to "Career".
    f. Showing four fingers will set the mode to “Love”.
    g. Showing all five fingers will set the mode to “Yes/No”.
3. Google-Text-To-Speech (GTTS) for voiceover
    a. In order to make Gretchen seem more human, we will be using GTTS to read the predictions out loud.
    b. This also acts as a way to make the readings more accessible, especially to those who may be visually impaired.
4. PyGame, PIL for User Interface
    a. The UI has been designed using PyGame and PIL. This way, we are able to customize the entire layout of the application to our liking.
5. AI agent for more complicated predictions
    a. For Modes 2, 5, only 1 card is required for the prediction to be made. Therefore, we created a database in order to give the predictions for these two modes.
    b. For Modes 3, 4, 3 cards are required for the predictions to be made. Creating a database for all the possible combinations of 3 cards will be a tedious and time consuming process as the tarot deck has 78 cards. Hence, we decided to use an AI agent (https://openrouter.ai/google/gemma-4-26b-a4b-it:free) to give the prediction so that there will be a reading for any combination of 3 cards.

Implementation:

1. When the user opens the application, they will be able to view the homescreen. Based on which mode they would like to use, they need to show the appropriate number of fingers.
2. After the mode has been chosen, Gretchen will look down. This indicates that Gretchen is ready to analyse the cards and this is a sign for the user to proceed with drawing the required number of cards.
3. The user will draw the required number of cards from the deck and place it on the table within the field of Gretchen's sight.
4. Gretchen will analyse the card, look up and the relevant music will start playing before the prediction has been delivered.
5. After the analysis, the prediction will be delivered to the reader with text as well as text to speech.

Results:

The model successfully recognizes all of the cards in the deck after the training on our own dataset, which proves the effectiveness of the work we have done.

The model has been proven successful as it meets all the following criterias:

1. The model is able to recognise the gesture of our hands (Numbers 0-5) in order to set the mode of prediction as chosen by the user.
2. The program is able to analyse the card that has been presented accurately. For more complicated predictions, the program is able to also analyse 3 cards at once with high levels of accuracy showing that it can be trusted in delivering the right predictions to the user.
3. The implemented AI agent is able to deliver concrete predictions when presented with 3 cards.
4. The user is fully immersed throughout their entire interaction with Gretchen through it's movements, music and text-to-speech. This creates a similar atmosphere as when one visits a fortune teller.

Discussion:

Training the model proved to be the most challenging phase of the project. Preparing the dataset required a lot of time and effort as we needed to orient the card in different ways in order to ensure that the model will be able to recognize it in different situations.

Initially, we decided to use modes that require only one card. This was fairly easy to implement as we were able to come up with the text files by ourselves as each prediction is linked to each specific card. However, we still felt that we would be able to do something even better. Therefore, we decided to come up with modes that require 3 cards as a form of making an advance prediction. Creating our own text files for every combination of 3 cards will be quite time consuming. Therefore, we decided to use an AI agent (https://openrouter.ai/google/gemma-4-26b-a4b-it:free) to generate the prediction. This was slightly challenging at first as we needed to find a free AI agent that would work seamlessly with the rest of our project and truly be able to analyse the cards and give an accurate prediction. After trying out various agents, we were satisfied with the performance of gemma.

Upon further research into Tarot, we learnt that each tarot deck is unique to the owner and that it contains the energy of the owner. Since the tarot deck belongs to Gretchen and this particular Gretchen has been created by us, we decided to leave a piece of ourselves in the project to give it our personal touch. We decided to add a mysterious secret mode that the users can activate by showing their index finger (Number 1). The screen will immediately play the song “Never Gonna Give You Up” by Rick Astley, a favourite song of ours.

Conclusion:

In order to further expand the scope of our project, we would love to see it implemented in public spaces so that it will be more accessible to everybody. In order to do so, we would also like to make use of virtual cards in order to lower manufacturing costs as well as make it easier and more interactive for the user as well. By making tarot more accessible to the general public, more people are given the avenue to broaden their horizons, to be more open-minded, and learn about this art form that has captivated humanity for generations.